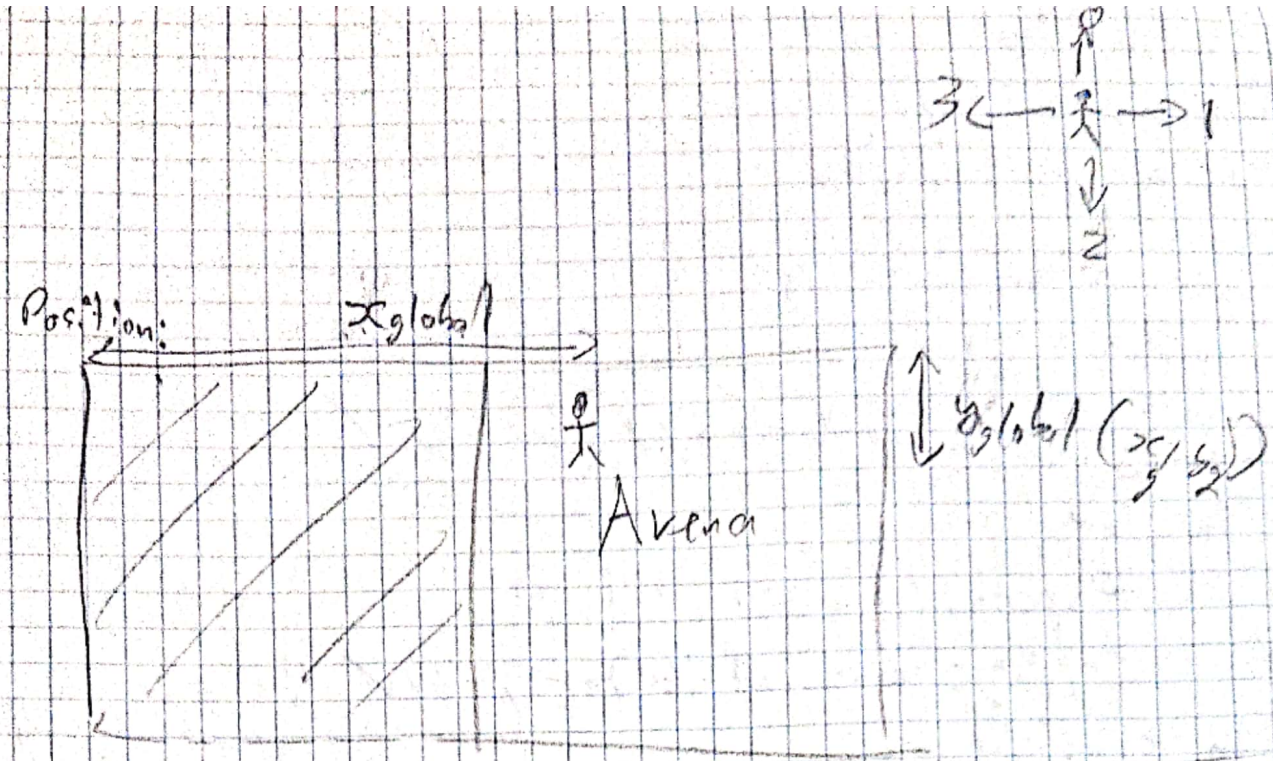
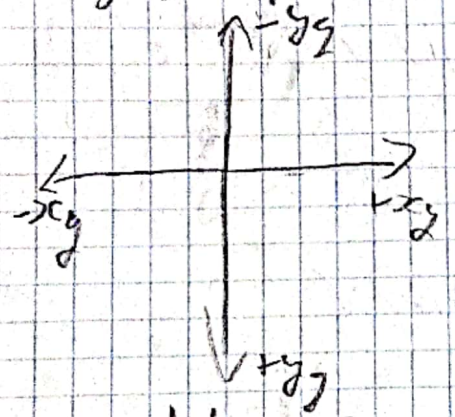


For global position:



Mapping:

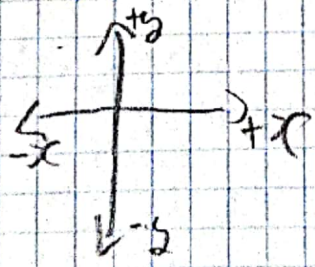
glob → 0: $(x_{glob}, -y_{glob})$

glob → 1: (y_{glob}, x_{glob})

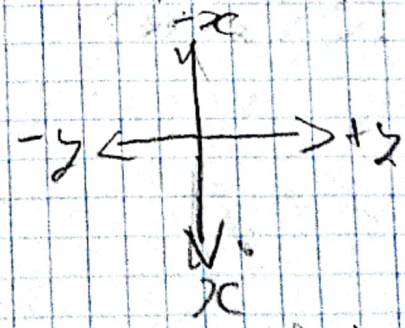
glob → 2: $(-x_{glob}, y_{glob})$

glob → 3: $(-y_{glob}, x_{glob})$

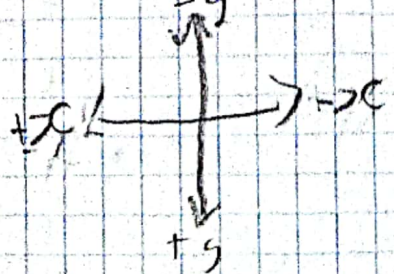
For orientation 0:



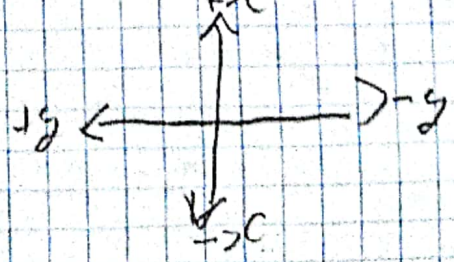
For orientation 1:



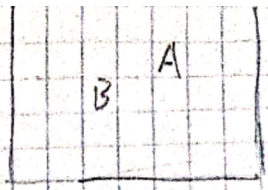
For orientation 2:



For orientation 3:



Testing:



(A) our pos: (16, 3)
(B) other pos: (14, 4)

In orientation 0:
Expect: (-2, -1)
Mapping: (16-16, -4+3) = (-2, -1) ✓

In orientation 1:
Expect: (1, -2)
Mapping: (-4-3, 14-16) = (1, -2) ✓

In orientation 2:
Expect (2, 1)
Mapping: (-14+16, 4-3) = (2, 1) ✓

In orientation 3:
Expect (-1, 2)
Mapping: (-4+3, -14+16) = (-1, 2) ✓

Rotations:

$${}^A P = {}^A_W R \, {}^W P$$

$$R_x(\gamma) = \begin{bmatrix} 1 & 0 \\ 0 & \cos\gamma \end{bmatrix}$$

$$R_z(\alpha) = \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix} \qquad R_y(\beta) = \begin{bmatrix} \cos\beta & 0 \\ 0 & 1 \end{bmatrix}$$

For W → Orientation 0: 1. $\alpha = 180°$ around Z
2. $\beta = 180°$ around Y

$${}^S_W R = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

$${}^O_S R = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

$${}^O P = {}^O_S R \, {}^S_W R \, {}^W P$$

$$= \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} {}^W P = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} {}^W P$$

Then ${}^O P_A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 16 \\ 3 \end{bmatrix} = \begin{bmatrix} 16 \\ -3 \end{bmatrix}$, ${}^O P_B = \begin{bmatrix} 14 \\ -4 \end{bmatrix}$

${}^O P_B - {}^O P_A = \begin{bmatrix} -2 \\ -1 \end{bmatrix}$ ✓

For W → Orientation 1: 1. $\alpha = 270°$ around Z
2. $\gamma = 180°$ around X

$${}^1 P = {}^1_S R \, {}^S_W R \, {}^W P$$

$${}^S_W R = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \qquad {}^1_S R = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$${}^1 P = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} {}^W P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} {}^W P, \quad {}^1 P_A = \begin{bmatrix} 3 \\ 16 \end{bmatrix}, \quad {}^1 P_B = \begin{bmatrix} 4 \\ 14 \end{bmatrix}$$

${}^1 P_B - {}^1 P_A = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$ ✓

CamScanner

For Orientation W->Z : 1. $\beta = 180^\circ$ around Y

$^zP = {}^z_wR\, {}^wP$, ${}^z_wR = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$

${}^zP_A = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 16 \\ 3 \end{bmatrix} = \begin{bmatrix} -16 \\ 3 \end{bmatrix}$, ${}^zP_B = \begin{bmatrix} -14 \\ 4 \end{bmatrix}$

${}^zP_B - {}^zP_A = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ ✓

For W -> Orientation 3 : 1. $\alpha = 270^\circ$ around Z
2. $\beta = 180^\circ$ around Y

${}^3P = {}^3_5R\, {}^5_wR\, {}^wP$

$= \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} {}^wP = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} {}^wP$

${}^3P_A = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}\begin{bmatrix} 16 \\ 3 \end{bmatrix} = \begin{bmatrix} -3 \\ -16 \end{bmatrix}$, ${}^3P_B = \begin{bmatrix} -4 \\ -14 \end{bmatrix}$

${}^3P_B - {}^3P_A = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$ ✓

CamScanner